

# 31338 Network Servers

## Lab 5a: Managing Users and Groups in CentOS

### Aims

1. Add, modify and delete users and groups
2. Understand how the actions are performed and how to configure users and groups

### Task 1: Understanding the Linux Identity Files

CentOS stores the account information in the file */etc/passwd*. Use the *cat* command to open it in the terminal. In this file, each row records an account. Specifically, see the root user's line:

```
root:x:0:0:Super User:/root:/bin/bash
```

These items are delimited by colons (:). Can you tell what do they mean?

The passwords attached to the accounts are stored in */etc/shadow*. The format of each row is like the account file *passwd*. Have a look at this file. Notice that many rows have the exclamation mark following the account name. What does it mean?

Similar to the user account, CentOS stores the group information in the file */etc/group*, and group permissions in the file */etc/gshadow*.

When creating a new user account, some default settings are copied to it. For example, the shell environment, a README file in the home directory. The root user can configure them as a skeleton. This is achieved by setting the contents in the directory */etc/skel*.

### Task 2: Managing Users and Groups with the Command Line

Use the command *useradd* to create a new user account peter with the full name "Peter Griffin". Then use the command *passwd* to set the password for peter to login. Check the peter account in the last line of the file */etc/passwd*.

You can use either *ssh* or *su* to enter peter's account

```
ssh peter@localhost
```

or

```
su peter
```

Then, exit the user's account simply by typing *exit*.

Use the command *groupadd* to create a new group 'feit', then add the user 'peter' to this group.

Sometimes, you need to manually configure the user account, e.g., when the user does not have the home directory, you need to manually configure it. Create a new user louis by using the argument *-M* in *useradd*. In this case, you need to setup the environment to ensure the user louis can login correctly. Before the manual configuration, can you use the username and password of louis to login?

Copy the whole directory */etc/skel* to the louis' home directory. Another issue is about the permission. Note that the contents are transferred by the root user, you need to use the *chown* and *chmod* commands to change it to louis.

## Task 3: Account Security – Locking and Expiration

You can use the command *usermod* command to modify a user account. Use *man usermod* to see how to use this command.

Before doing any modification, check peter's line in the shadow file */etc/shadow*. If peter's password has been setup, you will see a long hash string after 'peter:'.

Execute the following command to lock peter's account

```
usermod -L peter
```

Then check the shadow file in the peter's row. Can you still login peter's account using ssh? If not, how to unlock peter's account?

Hint: either edit the shadow file or use the usermod command.

Next, set up an expiration date using

```
usermod -e <YYYY-MM-DD> <user>
```

## Task 4: Safe Editing with vipw and vigr

It is strongly unrecommended to directly use a text editor such as vim to edit the following files, */etc/passwd*, */etc/shadow*, */etc/group* and */etc/gshadow*. (Why?) Instead, use *vipw* and *vigr* to edit these configurations.

Can you use the two commands to add and modify the user account and group?

## Task 5: Graphical User Management

You can use the graphical Users GUI in the Settings app to manage users. Try to use GUI to add a new user 'steve'. What details cannot you see or set via the GUI?

Record notes in your learning journal.